

Rubber Sheet & Ball

Sheet height $\propto -\Sigma m/r$; ball rolls along $-\nabla h$ — embedding metaphor, not GR.



About this model

Imagine a flexible rubber sheet stretched flat. When massive objects are placed on it, they create depressions, much like a bowling ball placed on a trampoline. This simulator visualizes a classic embedding diagram, where the height of the sheet, h , at any point is proportional to the negative sum of the gravitational potentials from each mass: $h \propto -\sum (G m_i / r_i)$. Here, G is the gravitational constant, m_i is a mass, and r_i is the distance from that mass. The resulting landscape is a direct, visual map of the gravitational potential field. A small ball placed on this contoured sheet will roll downhill. Its motion is governed by the local slope, meaning it accelerates in the direction of the steepest descent, which is the negative gradient of the height: $a \propto -\nabla h$. This rolling ball approximates the trajectory of a test mass moving under the

influence of gravity in a two-dimensional plane, following Newton's laws of motion and universal gravitation. The model simplifies real three-dimensional gravity by confining motion to a two-dimensional surface and using height as an analog for potential energy. It is a pedagogical tool, not a general relativity simulation; the curvature is a visualization aid, not spacetime itself. By interacting, students learn to connect the abstract concept of a scalar potential field to a tangible landscape, see how forces arise from gradients, and predict object motion by reading the shape of the terrain.

Who it's for: High school and introductory college physics students learning about Newtonian gravity, potential energy, and vector fields, as well as educators seeking a visual metaphor for gravitational fields.

KEY TERMS

- Gravitational Potential
- Gradient
- Potential Energy
- Newton's Law of Universal Gravitation
- Force Field
- Equipotential Line
- Test Mass
- Embedding Diagram

How it works

Museum-style gravity well: height mimics Newtonian potential; the marble finds the valley.

Frequently asked questions

- #### Is this a simulation of Einstein's general relativity?

No. While the 'rubber sheet' analogy is often used to popularize the idea of curved spacetime in GR, this simulator explicitly models Newtonian gravity. The sheet's curvature is a visual representation of gravitational potential energy, not spacetime geometry. The ball's motion is calculated from the slope (the force vector), illustrating $F = -\nabla U$, a core Newtonian concept.
- #### Why does the ball sometimes orbit and sometimes fall in?

The ball's path depends on its initial speed and direction, analogous to a planet's orbital mechanics. With the right tangential velocity, it can enter a stable orbit where the inward pull (downhill slope) provides the necessary centripetal force. Too little speed, and it falls directly inward; too much, and it may escape the depression on a hyperbolic path. This demonstrates the conservation of energy in a gravitational field.
- #### What are the main simplifications or limitations of this model?

The model is confined to two dimensions, whereas real gravity acts in three. The 'mass' creating the depression does not itself move in response to the ball, ignoring Newton's third law. Friction on the sheet is typically neglected, so the ball's mechanical energy is conserved, unlike real satellites that experience atmospheric drag. It is a useful metaphor, not a precise computational tool.

How is the 'height' of the sheet related to real gravitational energy?

The height is directly proportional to the negative gravitational potential, U . Lower height means lower (more negative) potential energy. The ball rolls to minimize its potential energy, just as objects in gravity are attracted to regions of lower gravitational potential. The steeper the slope (larger gradient of h), the stronger the gravitational force pulling the ball downhill.

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NEW SCHOOL

1D Force Field

▶ Launch Simulator

$U(x)$ presets, $F = -U'$, bead on potential, phase (x,v) and $E(t)$.

NEW SCHOOL

Inclined Plane: Work & η

▶ Launch Simulator

$W = F \cdot s$, friction and gravity work, ΔU , efficiency $\eta = \Delta U / W_F$; slide-down preset.

NEW SCHOOL

Spring Work ($F-x$)

▶ Launch Simulator

$F = kx$ graph, shaded $W = \text{area} = \frac{1}{2}kx^2$; $U(x)$ curve vs Spring-Mass dynamics.

FEATURED SCHOOL

Force Diagram Builder

▶ Launch Simulator

Place objects, add force vectors, see net force and resulting acceleration.

SCHOOL

Atwood Machine

▶ Launch Simulator

Two masses on a pulley. Adjust masses to see acceleration and tension.

NEW SCHOOL

Arch & Wedge Statics

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Compressive thrust in a stone arch and wedge force resolution (schematic).